THE INNOVATOR’S COURT

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PROBLEM STATEMENT 3

Disasters break the network. Then they break the swarm's knowledge.

In a disaster zone the communication network is usually the first casualty and today's drone swarms are built on the assumption that it survives.

- Towers collapse, power fails, rubble and buildings block signals. Links drop for minutes at a time.
- Centralised control relies on a single ground station. Losing that link leaves drones stranded or re-searching covered ground.
- Who it affects: survivors in the first critical hours · NDRF and SDRF teams entering unmapped, hazardous ground · commanders forced to decide on stale information.

But the harder problem comes next

When the network breaks, the swarm partitions it splits into groups that can no longer talk to each other. Both groups keep flying and keep observing, so you end up with two half-pictures of the same disaster.

GROUP A · D1, D2	GROUP B · D3
Survivor at Zone X	Survivor at Zone X
confidence 0.72	**confidence 0.76**

Confidence = how sure the drone is, on a 0–1 scale. 0.72 means "fairly sure, not certain."

One person seen twice — or two people?

Add the numbers and you get a confident answer that may be completely wrong. Send a team to a survivor who was already found, and someone else waits.

WHAT JEEVANMESH IS

JeevanMesh is a disaster-response drone swarm with no central controller. Every drone carries the mission rules, decides for itself, records what it sees as verifiable evidence, and can rebuild one shared, trusted picture with the others after any break in communication.

How a mission actually runs, end to end

1 Before take-off

Pre-flight goals and external data form a mission rulebook. It is copied to every drone, allowing for autonomous operation if the commander becomes unreachable.

2 In the air

Drones score areas based on potential learning, searching in adaptive spirals that tighten as evidence builds. No lawnmower sweeps.

3 On a detection

Drones log detailed evidence objects rather than basic alerts. All findings require multi-sensor or multi-drone confirmation before being declared.

4 When something fails

Neighbours instantly take over tasks. Weak links self-repair, and low battery triggers a safe handover.

5 When the network dies

The swarm splits, works independently, and stores mission memory locally.

6 When the link returns →

Memories sync, duplicates merge, conflicts are flagged, and one trusted mission picture is rebuilt.

01 · SEARCH

Spend scarce battery where information is worth the most.

02 · REMEMBER

Keep every finding verifiable, and merge separated memories correctly.

03 · RECOVER

Never let a lost drone or a lost link take unfinished work with it.

USE CASES

Floods · earthquakes · landslides · wildfires · cyclones · urban collapse · industrial accidents · night and low-visibility conditions

THE ARCHITECTURE

Human commander

Sets mission goals and constraints, approves high-risk actions, monitors the swarm and intervenes when needed.

Sensing & perception

RGB + thermal cameras, audio, RF/phone scanner, IMU/GPS and air/smoke/gas sensors convert raw data into structured findings onboard.

JeevanCore

- Shared protocol + onboard decision engine
- Converts mission goals into swarm rules
- Builds global situational awareness
- Assigns & changes drone roles
- Manages safety, geofencing & risk
- Resolves conflicts, re-checks & final decisions
- Not a central server every drone runs it

External systems

Connects NDRF/SDRF and incident command with satellite, weather, maps/building data and emergency-service feeds.

Communication layer

Peer-to-peer mesh with store-carry-forward, adaptive routing, relay repositioning and link-quality estimation for disrupted networks.

Mission intent & safety policy

Mission goals + safety rules: geofencing, no-fly zones, DGCA limits and emergency protocols stored on every drone.

JeevanSearch · JeevanMemory · JeevanMesh

Three core intelligence layers: where to search, what evidence to trust, and how tasks are redistributed when drones or links fail.

The rule that makes it decentralised

All actions are checked against local rules: routine tasks execute instantly, while high-risk actions require approval. If disconnected, drones run safe fallbacks and pause risky actions until the link returns—never guessing or freezing.

JEEVANMESH

SELF-HEALING RESCUE SWARM

One Mission. One Swarm. No Single Point of Failure.

Search Intelligently. Remember Consistently. Recover Automatically.

Key Actors

Human Commander · Drone Swarm · External Systems

MISSION WORKFLOW

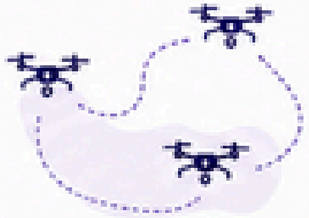
1 MISSION SETUP

- Commander sets goals and constraints
- Swarm takes off and syncs initial plan



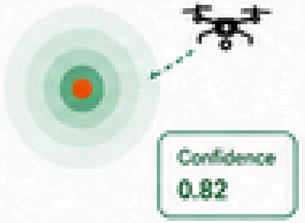
2 DEPLOY & EXPLORE

- Drones spread out using adaptive search (whale-inspired)
- Data captured using multi-sensors



3 DETECT & EVALUATE

- Potential survivors/ areas detected
- Confidence scored and shared



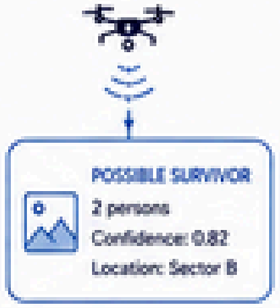
4 DECIDE & ASSIGN

- Best drone(s) assigned based on role, battery, network & capability
- Roles adapt dynamically



5 EXECUTE & SHARE

- Drones execute tasks
- Share "meaning before media" when bandwidth is low



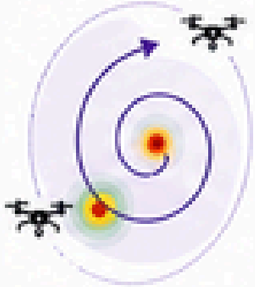
6 ADAPT & RECOVER

- If failure/partition occurs, swarm adapts
- Mission continues with remaining drones



ADAPTIVE SEARCH (Whale-Inspired Spiral)

Start broad.
Focus on high-probability zones.
Tighten the spiral.
Confirm & move on.



WHEN FAILURE HAPPENS (Any Link, Drone, GPS, Battery, or Network)

Detect — Swarm detects an issue

Isolate — Issue isolated to prevent spread

Reassign — Tasks handed over to capable drones

Continue — Mission continues locally

Reconnect — Groups reconnect when possible

Merge & Sync — Data merged, conflicts resolved, state updated

WHAT MAKES JEEVANMESH SELF-HEALING

Decentralized Control
No single point of failure

Partition-Tolerant Memory
Mission state survives even when split

Adaptive Coordination
Roles, routes & priorities change in real-time

Resilient Communication
Works with or without infrastructure

Safe & Reliable
Safety-first policies, human in control

PILLAR 1 : SEARCH

The perception pipeline, running on every drone

Sensors → sensor fusion → object detection → evidence reasoning → local world state → navigation/search → mission decision → flight control

THE FIVE SEARCH STAGES

Broad spiral → gather evidence → confidence rises → tighten around target → second-drone verification

- Broad spiral: Explore wide → gather evidence → tighten → verify.
- Utility field U(x,y,t): Scores best search areas using confidence, battery, obstacles, comms & weather.
- Whale-inspired search: Spiral shrinks as evidence strengthens.
- Two-step confirmation: Second drone/sensor verifies every detection.



Swarm prioritizes information by mission need: CRITICAL (survivors) → HIGH (safe routes) → MEDIUM (infrastructure) → LOW (mapping).

PILLAR 2 : REMEMBER

An observation is a record, not a flag

Not survivor = true, but an evidence object: observation ID · which drone · 3D position · time · sensor type · confidence · evidence reference · mission context. That history is called provenance, and it is what makes correct merging possible later.

SIX MEMORY SERVICES

- **provenance tracking**
- **freshness & decay**
- **deduplication & matching**
- **conflict detection**
- **fusion & reconciliation**
- **state versioning**

- **Freshness & decay:** Old evidence automatically loses weight. Mission updates are versioned, ensuring drones can always distinguish new data from old.
- **Knowledge protection:** The swarm identifies "Single Points of Knowledge." Critical info held by only one drone is copied to a neighbor before it disconnects.

The matching question: same event, or two events?

Decided by three checks: spatial distance, time gap, and whether both sensors could be seeing the same event. Borderline cases stay separate and trigger verification

Principle	What It Prevents
Event IDs — every observation and task change carries a unique identifier	The same finding arriving by three routes being counted as three findings
Commutative merges — memories combine to the same result whichever way groups reconnect	Two halves ending up with different final pictures
Evidence weighting — confidence combines with a discount for how related two observations are	Fake certainty. Independent sensors add a lot; a repeat adds almost nothing; the total can never exceed 1
Quorum for critical claims — no single drone can declare a survivor alone	One faulty sensor sending a rescue team to an empty building
Conflict as first-class state — disagreement is stored, never averaged away	A clean-looking answer that hides an unresolved question
Eventual consistency — groups are allowed to diverge, on the guarantee that they converge	Freezing the mission until everyone agrees

PILLAR 3 : RECOVER

When a drone or a link disappears, its mission doesn't

THE EIGHT RECOVRY STAGES

- Dynamic roles: Roles (scout, relay, etc.) are fluid and assigned instantly based on the mesh network's needs.
- Health monitoring: Drones send regular heartbeats; missing signals or faulty sensors trigger immediate isolation.
- Task reassignment: A missing drone's tasks and context are locally reassigned based on distance, battery, and urgency.
- Battery-aware handoff: Low-battery drones pass on tasks and critical evidence before returning safely.
- Route replanning: Hazards trigger on then fly route replanning and swarm-wide warnings.
- Physical data relay: If disconnected, drones physically carry data to an area with an active connection.

Failure → Response

Drone failure	Tasks re-bid, winner inherits task and context
Link failure	Reroute, then reposition a relay, then carry the data physically
GPS loss	Navigate by camera, motion sensors and distance to neighbours
Battery depletion	Hand over context, push critical evidence, return to base
Blocked route	Store the hazard as evidence, replan, warn the others
Sensor fault	Demote the role, exclude from verification
Swarm partition	Both halves keep flying under the same policy, reconcile on reconnect
Loss of internet	Nothing changes — we never relied on it

WIREFRAME


JEEVANMESH
Self-Healing Rescue Swarm
Mission Time
01:28:47
MISSION ACTIVE
Area Covered
18.6 / 40.0
acres
Survivors Detected
3 suspected
1 verified
Mission Time
01:28:47
Est. Remaining
00:41:13
Energy Avg.
64%
All Drones
Network Health
94%
Good
Active Drones
3 / 4
1 offline
Live Rescue Map
Heatmap
Routes
Obstacles
2D
3D
D1
72%
82%
D3
82%
28%
61%
D2
34%
D4
OFFLINE
100 m
Drone Status
D1 – SCOUT
Alt: 63 m | Spd: 12 m/s
Task: Search Sector A
72%
Good
D2 – VERIFY
Alt: 58 m | Spd: 10 m/s
Task: Verify Hotspot #1
34%
Weak
D3 – RELAY
Alt: 81 m | Spd: 11 m/s
Task: Relay & Scan
82%
Good
D4 – SUPPLY
Last seen: 01:15:32
Status: Offline
0%
No Link
Search Confidence
Low
High
Top Hotspots
82%
Grid (428, 711)
Updated 01:28:12
61%
Grid (255, 344)
Updated 01:27:45
28%
Grid (612, 198)
Updated 01:26:51
Network Health
100%
50%
0%
00:00
00:45
01:30
94%
Good
Bandwidth
2.4 Mbps / 10 Mbps
Partitions
2
Alerts & Events
01:28 New evidence added at Hotspot #1
01:27 Acting as Relay for Sector A
01:26 Battery low – task reassignment prepared
01:22 Route blocked near (512, 640)
01:18 Possible survivor detected (RGB)
Legend
Drone
Active Link
Weak Link
No Link
Search Path (Spiral)
Planned Route
Obstacle / Blocked

JeevanMesh

Self-Healing Rescue Swarm

Self-Healing — No Single Point of Failure
Scalable — 3 or 30+ Drones
Resilient — Works in Disruptions
Mission-Driven — Evidence First, Lives First

MISSION OVERVIEW

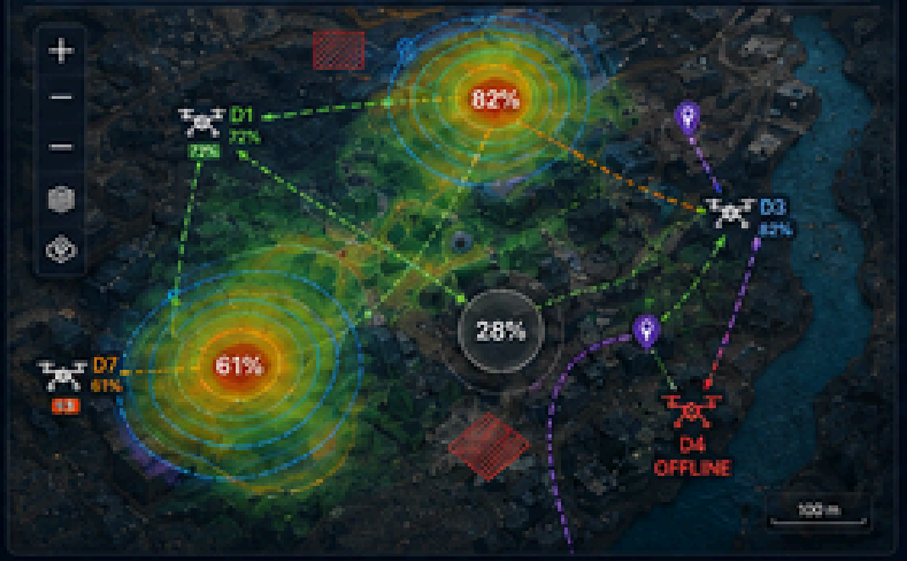
- Area Covered: 18.6 acres / 40.0 acres — 48%
- Survivors Detected: 3 — 1 Verified | 2 Suspected
- Active Drones: 3 / 4 — 1 Offline
- Energy (Avg.): 64%
- Network Health: 94% — Good

RECENT ALERTS

- Low battery on D2 (34%) — 00:51 ago
- D4 offline – last seen 01:15:32 — 02:13 ago
- Route blocked ahead — 03:21 ago

LIVE RESCUE MAP

Heatmap | Links | Obstacles | Routes | All Layers



DRONE STATUS

View all →

- D1 - SCOUT — Alt: 63 m | Spd: 12 m/s — Task: Search Sector A — 72% — Good
- D2 - VERIFY — Alt: 58 m | Spd: 10 m/s — Task: Verify Hotspot A — 34% — Weak
- D3 - RELAY — Alt: 81 m | Spd: 11 m/s — Task: Relay & Scan — 82% — Good
- D4 - SUPPLY — Last seen: 01:15:32 — Status: Offline — 0% — No Link

NETWORK HEALTH

Mesh Connectivity



Link Quality



WEATHER

26°C
Light Rain
Wind: 12 km/h
Humidity: 78%

SEARCH & CONFIDENCE

Probability Heatmap
Low — High

Top Hotspots

- 82% Grid (148, 71)
- 61% Grid (532, 304)
- 28% Grid (418, 196)

TASKS & EVENTS

View all →

- 01:28 New evidence added at Hotspot #1
- 01:24 Acting as Relay for Sector A
- 01:20 Battery low – task reassignment prepared
- 01:18 Route blocked near (532, 304)
- 01:16 Connection lost – rerouting locally
- 01:11 Possible survivor detected (S03)



EVIDENCE & ALERTS

View all →

Critical Evidence



Hotspot #1
Possible survivor
Time: 01:21:55

Active Alerts

- Low battery on D2 (34%)
- D4 offline – last seen 01:15:32
- Route blocked ahead

MOBILE OPERATOR VIEW





- **Live Map** — Real-time awareness
- **Evidence** — Meaning before media
- **Alerts** — Critical notifications
- **Offline First** — Works without network

JEEVANMESH SYSTEM FLOW

1. **PERCEIVE** — Collect multi-sensor data — Sensors
2. **UNDERSTAND** — Detect & interpret intelligently — YOLO / VLM
3. **LOCAL MEMORY** — Store as evidence object — ID • Time • Location • Type • Confidence • Provenance
4. **SHARE** — Exchange via mesh, Store-and-carry when needed
5. **CONSISTENCY** — Merge, deduplicate, resolve conflicts
6. **DECIDE & ASSIGN** — Reassign tasks, change roles
7. **ACT** — Execute mission, adapt & search

FAILURE / PARTITION — Drone fails, link breaks, GPS lost, battery low

SELF-HEALING RECOVERY — Reroute, reassign, store-carry-forward, reconnect & continue

Continuous Loop: Search Intelligently → Remember Consistently → Recover Automatically → Save More Lives

ARE THEY THE SAME?

Flowchart
Shows the step-by-step flow of a process or system.
Example: Evacuation plan, data flow.

Architecture Diagram
Shows the high-level structure and components of a system and how they connect.
Example: Software architecture, network layers.

Concept Visual
Shows the big idea or vision in an intuitive, visual and inspiring way.
Example: How JeevanMesh works in the real world.

IN SHORT

They are related but not the same.

Flowchart = How it flows (steps)

Architecture = What it is (structure)

Concept Visual = Why it matters (idea/impact)

SCALABILITY/FUTURE IMPACT

Scaaibility

- Scales by adding units, not servers: Whether 3 drones or 30, the same coordination rules apply. Main limits are radio bandwidth and safe aircraft separation.
- Works across autonomous machines: The same system can coordinate drones, ground robots, boats, and fixed sensors.
- Useful beyond disaster response: Similar coordination is needed in agriculture, logistics, infrastructure inspection, and surveying.
- Long-term vision: Build a shared mission-memory layer where autonomous teams share knowledge and decisions, just as they already share maps.

JeevanLink : the emergency network for people

The same mesh that carries evidence can carry survivors' voices. Victim SOS beacons · emergency messages and broadcasts · location approximation for people who cannot be seen · offline connectivity when the ground network is gone.

Phones, people, drones and responders on one mesh when every other network has failed.

The Outcomes

More survivors found · faster response · lower risk to rescuers · efficient resource use, with no drone-hour spent re-discovering something the swarm already knew.